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# PERFORMING HIV-1 RNA QUANTITATIVE TEST USING CAP/CTM96

## DOCUMENT REVISION HISTORY

| Revision No. | Changes | Reviewer | Date |
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## 1. PURPOSE

To describe the procedure for quantitative determination of HIV-1 from plasma using the COBAS AmpliPrep/COBAS TaqMan HIV-1 Test v2.0 for monitoring of HIV-1 infection.

## 2. PRINCIPLE

The COBAS AmpliPrep/COBAS TaqMan HIV-1 Test is a nucleic acid amplification test for the quantitation of Human Immunodeficiency Virus Type 1 (HIV-1) RNA in human plasma.

The test is based on three major processes:

1. Specimen preparation to isolate HIV-1 RNA;
2. Reverse transcription of the target RNA to generate complementary DNA (cDNA), and
3. Simultaneous PCR amplification of target cDNA, HIV-1 Quantitation Standard (QS) Armored RNA and detection of cleaved dual-labeled oligonucleotide probe specific to the target.

The Master Mix reagent contains primers and probes specific for both HIV-1 RNA and HIV-1 QS RNA. The detection of amplified DNA is performed using a target-specific and a QS-specific dual-labeled oligonucleotide probe that permit independent identification of HIV-1 amplicon and HIV-1 QS amplicon. The quantitation of HIV-1 viral RNA is performed using the HIV-1 QS. It compensates for effects of inhibition and controls the preparation and amplification processes, allowing a more accurate quantitation of HIV-1 RNA in each specimen. The HIV-1 QS is a non-infectious Armored RNA construct that contains HIV sequences with identical primer binding sites as the HIV-1 target RNA and a unique probe binding region that allows HIV-1 QS amplicon to be distinguished from HIV-1 target amplicon. The HIV-1 QS is added to each specimen at a known copy number and is carried through the specimen preparation, reverse transcription, PCR amplification and detection steps of cleaved dual-labeled oligonucleotide detection probes. The COBAS TaqMan Analyzer or COBAS TaqMan 48 Analyzer calculates the HIV-1 RNA concentration in the test specimens by comparing the HIV-1 signal to the HIV-1 QS signal for each specimen and control.

## 3. PERFORMANCE CHARACTERISTICS

Refer to COBAS® AmpliPrep/COBAS® TaqMan® HIV-1 Test, version 2.0 package insert (PCRL-EXT-001) for performance characteristics.

## 4. SPECIMEN REQUIREMENTS

Minimum of 1.00 to 1.1 ml of plasma extracted from fresh EDTA anticoagulated whole blood.

## 5. PATIENT PREPARATION

N/A

## 6. SPECIMEN STORAGE

Specimens are viable if stored at;

| Temperature | Period |
|---|---|
| 25 to 30◦C (room temperature) | 1 Day |
| 2 to 8◦C | Up to 6 Days |
| -20 to -80 | Up to 6 weeks |

Plasma specimens may be frozen and thawed up to five times.

## 7. PERSONAL PROTECTIVE EQUIPMENT

a. Lab Coat
b. Powder-free disposable gloves
c. Closed toed shoes

## 8. ENVIRONMENTAL AND SAFETY CONTROLS

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| Follow standard laboratory practices while working with biohazardous material. |
| Keep the main covers and the loading panes closed and in place while the instruments are operating. |
| When working with open main cover while the instrument is powered on, always put the instrument in maintenance mode or in shutdown status first. |
| Be sure to wear appropriate protective equipment, including but not limited to, safety glasses, laboratory coats and disposable gloves. |
| If any biohazardous material is spilled, wipe it up immediately and apply disinfectant. |
| If a sample or waste solution comes into contact with your skin, immediately apply disinfectant, and wash it off with soap and water. |
| When handling with syringes and reagent tips during maintenance actions, be sure to wear puncture resistance hand protection to prevent possible hand injury. |
| Do not stare into the laser transmitters beam as your eyesight maybe be severely damaged. |

## 9. REQUIREMENTS

| Reagents | Supplies | Equipment |
|---|---|---|
| CAP/CTM HIV-1 Test kit v2.0<br>• HIV-1 CS1<br>• HIV-1 CS2<br>• HIV-1 CS3<br>• HIV-1 CS4<br>• HIV-1 (–) C<br>• HIV-1 L(+)C<br>• HIV-1 H(+)C | Sample Processing Units (SPU)<br>Input Sample tubes (S-Tubes)<br>K-Tubes<br>K-Tips<br>CAP/CTM Laboratory consumables bundle<br>• Sterile filter pipette tips<br>• Powder free glove<br>• Biohazard bag autoclavable<br>• Lab coats | Cobas Ampliprep/Cobas Taqman 96<br>Vortex mixer<br>Biosafety cabinet |

| | | |
|---|---|---|
| | • Laboratory marker<br>SPU racks<br>Reagent racks<br>100-1000µl Micro pipettes | |

10. **CALIBRATION PROCEDURE**

To be performed by service engineer in line with recommendations of the manufacturers service manual once per year. A certificate of performance will be issued and retained in the laboratory.

**11. QUALITY CONTROL PROCEDURE**

| Step | Action |
|---|---|
| **1.** | Run one replicate of the COBAS TaqMan Negative Control, the HIV-1 Low Positive Control and the HIV-1 High Positive Control on each test batch. |
| **2.** | There are no requirements regarding the position of the controls on the sample rack. |
| **3.** | The batch is valid if no flags appear on the HIV-1 H(+)C, (**HIV-1 L(+)C, and CTM (–) C**) controls. |

**12. PROCEDURE STEP BY STEP**

12.1. COBAS AmpliPrep and Cobas TaQman Instrument Set-up

| Step | Action |
|---|---|
| 1. | Turn the Data Station for the AMPLILINK software ON. |
| 2. | Log onto the Windows XP operating system. |
| 3. | Double click the AMPLILINK software icon. |
| 4. | Log onto AMPLILINK software by entering the assigned User ID and password. |
| 5. | Check the supply of PG WR using the Status Screen and replace if necessary. |
| 6. | Perform all Maintenance that is listed in the Due Tab of the COBAS AmpliPrep and Cobas TaQman Instruments. Cobas Ampliprep will automatically prime the system. Refer to Cobas Ampliprep maintenance procedure (PCRL-TECH-020) and Cobas Taqman maintenance Procedure (PCRL-TECH-021) for detailed maintenance steps. |

12.2. Loading of Reagent Cassettes

| Step | Action |
|---|---|
| 1. | Removed the reagents from 2-8°C storage, immediately loaded onto the COBAS AmpliPrep Instrument and allowed to equilibrate to ambient temperature for at least 30 minutes before the first sample is to be processed. |
| 2. | Place HIV-1 CS1 onto a reagent rack and load onto rack position A of the COBAS AmpliPrep Instrument. |
| 3. | Place HIV-1 CS2, HIV-1 CS3 and HIV-1 CS4 onto a separate reagent rack and load onto rack position B, C, D or E of the COBAS AmpliPrep Instrument. |

12.3. Loading of Disposables and K-carriers

| Step | Action |
|---|---|
| 1. | Determine the number of CAP/CTM reagent cassettes, SPUs, S-tubes, K-tips and K-tubes needed. |
| 2. | Place the SPUs in the SPU rack(s) and load the rack(s) onto rack position J, K or L of the COBAS AmpliPrep Instrument. |
| 3. | Load full K-tube rack(s) onto rack position M, N, O or P of the COBAS AmpliPrep Instrument. |
| 4. | Load full K-tip rack(s) onto rack position M, N, O or P of the COBAS AmpliPrep Instrument. |
| 5. | Load K-carriers on pack positions 5-12 of the Cobas TaQman instrument. |

12.4. Ordering and loading of specimens

| Step | Action |
|---|---|
| 1. | Prepare sample racks as follows: Attach a barcode label clip to each sample rack position where a specimen (S-tube) is to be placed. |
| 2. | Attach one of the specific barcode label clips for the controls [CTM (–) C, HIV-1 L(+)C, and HIV-1 H(+)C,] to each sample rack position where the controls (S-tube) are to be placed. |
| 3. | Place one Input S-tube into each position containing a barcode label clip. |
| 4. | Using the AMPLILINK software, create specimen orders for each specimen and control in the Orders Window Sample folder. |
| 5. | Select the test file HICAP96 and complete by saving. |
| 6. | Assign specimen and control orders to sample rack positions in the Orders Window Sample Rack folder. |
| 7. | Print the Sample Rack Order report to use as a worksheet. |
| 8. | Prepare specimen and control racks in the designated area for specimen and control. Vortex each specimen and control [CTM (–) C, HIV-1 L(+)C, and HIV-1 H(+)C] for 3 to 5 seconds. Avoid contaminating gloves when manipulating the specimens and controls. |
| 9. | Transfer 1000 to 1100 µL of each specimen and control [CTM (–) C, HIV-1 L(+)C, and HIV-1 H(+)C] to the appropriate barcode labeled Input S-tube using a micropipettor with an aerosol barrier or positive displacement RNase-free tip. *Avoid transferring particulates and/or fibrin clots from the original specimen to the Input S-tube.* |
| 10 | Transfer Specimens and controls to tube positions as assigned and record on the worksheet. Avoid contaminating the upper part of the S-tubes with specimens or controls. |
| 11 | Load the sample rack(s) filled with Input S-tubes onto rack positions F, G or H of the COBAS AmpliPrep Instrument |

12.5. Start of COBAS AmpliPrep Instrument Run

| Step | Action |
|---|---|
| 1. | Open the AmpliPrep Systems window. In the System folder start the run by clicking the "**Start**" button. |

12.6. End of COBAS AmpliPrep Instrument Run and Transfer to COBAS TaqMan Analyzer

| Step | Action |
|---|---|
| 1. | Check for flags or error messages. |
| 2. | Remove processed samples and controls from the COBAS AmpliPrep Instrument on either sample racks (for COBAS TaqMan Analyzer without docking station) |
| 3. | Remove waste from COBAS AmpliPrep Instrument. |

12.7. Amplification and Detection

| Step | Action |
|---|---|
| 1. | The COBAS TaqMan Analyzer will automatically start moving the K-tubes to a K-carrier if a docking station is used. It is not necessary to click "Start" again. |
| 2. | Load K-carrier racks containing processed controls/samples and working Master Mix onto the COBAS TaqMan Analyzer within 120 minutes after being processed. |
| 3. | Do not expose processed samples and controls to light after completion of sample and control preparation. DO NOT FREEZE or STORE processed samples and controls at 2-8°C. |

12.8. End of COBAS TaqMan Analyzer Run

| Step | Action |
|---|---|
| 1. | At the completion of the COBAS TaqMan Analyzer run, print Results Report. Check for flags or error messages in the Result Report. Sample results with flags and comments are interpreted as described under section 14 of this SOP. After acceptance, store data in archive. |
| 2. | Remove used K-tubes from the COBAS TaqMan Analyzer |

## 13. CALCULATION OF RESULTS

The COBAS TaqMan Analyzer automatically determines the HIV-1 RNA concentration expressed in copies (cp)/mL for the specimens and controls**.**

## 14. INTERPRETATION OF RESULTS

**Control results**

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| The CTM (–) C (Negative Control) must yield a "Target Not Detected" result or run will be invalid. |
| The assigned range for HIV-1 L(+)C and HIV-1 H(+)C is specific for each lot of reagents, and is provided on the COBAS® AmpliPrep / COBAS® TaqMan® HIV-1 Test reagent cassette barcodes. |

**Patient results**

| **Titer Result** | **Interpretation** |
|---|---|
| Target Not Detected | Ct value for HIV-1 above the limit for the assay or no Ct value for HIV-1 obtained. |
| < 2.0E+01 cp/mL | Calculated cp/mL are below the Limit of Detection of the assay. Report results as "HIV-1 RNA detected, less than 20 HIV-1 RNA cp/mL". |
| ≥ 2.0E+01 cp/mL and ≤ 1.00E+07 cp/mL | Calculated results greater than or equal to 20 cp/mL and less than or equal to 1.00E+07 cp/mL are within the Linear Range of the assay. |
| > 1.00E+07 cp/mL | Calculated cp/mL are above the range of the assay. Report results as "greater than 1.00E+07 HIV-1 RNA cp/mL". If quantitative results are desired, the original specimen should be diluted with HIV-1-negative human EDTA-plasma and the test repeated. Multiply the reported result by the dilution factor. |

## 15. REPORTABLE INTERVAL OF EXAMINATION RESULTS

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|---|---|
| Minimum cut off | 20cp/ml |
| Maximum cut off | 10,000,000cp/ml |

## 16. BIOLOGICAL REFERENCE RANGES CLINICAL DECISION VALUES

As per Zambia Consolidated Guidelines for Treatment and Prevention of HIV (PCRL-EXT-003)

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|---|---|
| <1000cp/ml | Viral Suppression |
| >1000cp/ml | Suspected Treatment failure |

## 17. ALERT/CRITICAL VALUES (where appropriate)

N/A

## 18. LIMITATIONS AND SOURCES OF ERROR (INTERFERENCES)

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|---|---|
| 1 | This test has been validated for use with only human plasma collected in EDTA anticoagulant. Testing of other specimen types may result in inaccurate results. |
| 2 | The performance of the COBAS AmpliPrep/COBAS TaqMan HIV-1 Test has neither been evaluated with specimens containing HIV-1 groups O and N, nor with specimens containing HIV2. |
| 3 | The presence of AmpErase enzyme in the COBAS AmpliPrep/COBAS TaqMan HIV-1 Master Mix reduces the risk of amplicon contamination. However, contamination from HIV-1 positive controls and clinical specimens can be avoided only by good laboratory practices and careful adherence to the procedures specified in this Package Insert. |
| 4 | This product can only be used with the COBAS AmpliPrep Instrument and the COBAS TaqMan Analyzer or COBAS TaqMan 48 Analyzer. |
| 5 | Though rare, mutations within the highly conserved region of the viral genome covered by this test, primers and/or probes may result in the under-quantification of or failure to detect the virus. |
| 6 | Detection of the HIV-1 RNA is dependent on the number of virus particles present in the specimen and maybe affected by specimen collection methods and patient factors such as age, presence of symptoms and stage of infection. Write factors that can cause interferences and cross reactions and therefore potentially lead to inaccurate results. |
| 7 | Do not expose PCR samples to direct sunlight |

## 19. WASTE DISPOSAL

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| 1 | Discard all used SPUs together with S-tubes into a plastic biohazard bag. Seal and secure the contents of the of the waste bag before disposal. |
| 2 | Empty the contents of the waste reservoir in the sink and flush the sink thoroughly with water. Rinse the waste reservoir with water. |
| 3 | Remove the waste container from the Cobas Taqman analyser. Seal and secure the contents K-Tube waste container before disposal. |
| 4 | Put a biohazard bag into an appropriately labeled waste bin. |
| 5 | Autoclave the waste bags at 121ºC for 1 hour and take them for incineration. |
| 6 | Environmental health office is responsible for waste incineration. |

## 20. RELATED DOCUMENTS

| Document number | Document Title |
|---|---|
| PCRL-TECH-020 | Cobas Ampliprep maintenance procedure |
| PCRL-TECH-021 | Cobas Taqman maintenance Procedure |
| PCRL-EXT-003 | Zambia Consolidated Guidelines for Treatment and Prevention of HIV. |

## 21. REFERENCES

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| COBAS AmpliPrep/COBAS TaqMan HIV-1 Test package insert. |
| Cobas AmpliPrep instrument manual |
| Cobas Taqman analyzer instrument manual |

## 22. APPENDICES

**N/A**

## 23. STAFF ACKNOWLEDGEMENT

I have read, understood and agree to follow the procedure as documented:

| No. | Name | NRC No. | Signature | Date |
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